

provide a way for an Administrator to revoke access to Nodes that have been explicitly excluded from an operational group (albeit after a period of time).

Epoch keys are generated, managed, and stored by an Administrator on a per-Fabric basis. Each key SHALL be a [random](#) value of length [CRYPTO_SYMMETRIC_KEY_LENGTH_BITS](#) bits.

```
EpochKey = Crypto_DRBG(len = CRYPTO_SYMMETRIC_KEY_LENGTH_BITS)
```

Each epoch key has associated with it a start time that denotes the time at which the key becomes active for use by transmitting Nodes. Epoch key start times are absolute UTC time in microseconds encoded using the [epoch-us](#) representation.

4.17.3.1. Using Epoch Keys

Nodes sending group messages SHALL use operational group keys that are derived from the *current epoch key* (specifically, the epoch key with the *latest* start time that is not in the future). Nodes that cannot reliably keep track of time calculate the *current epoch key* as described in [Section 4.17.3.4, “Epoch Key Rotation without Time Synchronization”](#).

Nodes receiving group messages SHALL accept the use of any key derived from one of the currently installed epoch keys. This requirement holds regardless of whether the start time for the key is in the future or the past. This means Nodes continue to accept communication secured under an epoch key until that key is withdrawn by explicitly deleting the key from a Node’s group state by the key distribution Administrator.

Note that there is no end time associated with an epoch key. An epoch key marked with the maximum start time SHALL be disabled and render the corresponding epoch key slot unused.

4.17.3.2. Managing Epoch Keys

The epoch keys are managed using the [Group Key Management Cluster](#). For every group key set published by the key distribution Administrator, there SHALL be at least 1 and at most 3 epoch keys in rotation. Key additions or updates are done using the [KeySetWrite](#) command.

Key updates are idempotent operations to ensure the Administrator is always the source of truth. An epoch key update SHALL order the keys from oldest to newest.

Any epoch key update MAY deliver a partial key set but SHALL include [EpochKey0](#) and MAY include [EpochKey1](#) and [EpochKey2](#). Any update of the key set, including a partial update, SHALL remove all previous keys in the set, however many were defined.

An Administrator MAY completely remove a group key set from a Node using the [KeySetRemove](#) command.

4.17.3.3. Epoch Key Rotation

The key distribution Administrator generates new epoch keys on a regular basis, giving each a unique id and adding them to the list of existing epoch keys within a group. The start time for each new epoch key is scheduled to occur after a configurable *key propagation interval*. The propagation